Brazilian School Census (Censo Escolar) ETL Pipeline

A Data Engineering Tutorial Report

1.0 Business Understanding

1.1 Problem Statement

Education is one of the most critical pillars of national development. Understanding the state of basic education across a large and diverse country like Brazil requires sophisticated data management and analytical capabilities. The Brazilian National Basic Education Census (Censo Escolar) is one of the largest educational data collection efforts in the world, gathering detailed information annually from millions of schools, students, teachers, and facilities across all 26 states and the Federal District.

However, the sheer volume and complexity of this data — spanning from 2010 to 2021 and containing hundreds of attributes per record — makes traditional data processing approaches inadequate for timely and meaningful analysis. Without a scalable pipeline, stakeholders such as education policymakers, school administrators, and researchers would struggle to derive accurate insights, potentially leading to poorly informed decisions about resource allocation, infrastructure investment, and educational reform.

1.2 Business Objectives

The primary objective of this project is to transform approximately 2.5 GB of raw annual CSV files covering the Brazilian Basic Education Census from 2010 to 2021 into a structured, query-optimised data warehouse using Apache Spark as the core processing engine. The specific objectives are:

- Design and implement a pipeline that extracts raw CSV data from the Brazilian Open Data portal
- Convert CSV files into Parquet format for improved query performance
- Transform the data into a Star Schema consisting of clearly defined dimension and fact tables
- Load the final structured data into a PostgreSQL relational database
- Enable business intelligence dashboards through Metabase for stakeholder visualisation

1.3 Stakeholders and Context

The primary stakeholders include the Brazilian Ministry of Education (MEC) and the National Institute for Educational Studies and Research (INEP), responsible for collecting and publishing Censo Escolar data annually. Secondary stakeholders include state and municipal education secretariats that rely on this data to plan school infrastructure, allocate teachers, and manage student enrolment. Policy researchers and data analysts requiring aggregated and clean datasets for longitudinal studies also benefit significantly from this pipeline.

2.0 Data Understanding

2.1 Dataset Overview

The dataset is sourced from the Brazilian National Basic Education Census (Censo Escolar), published annually by INEP and freely available through Brazil's open data portal. The dataset spans 2010 to 2021, producing twelve annual CSV files. Each file contains approximately 370 columns and around 200 MB of disk space, bringing the total raw dataset to approximately 2.5 GB. The data captures every registered basic education school in Brazil, including attributes related to school infrastructure, geographical location, student demographics, teacher qualifications, and availability of resources such as internet access, libraries, water supply, sanitation, and computer laboratories.

Attribute	Category	Data Type	Description
CO_UF	Geographical	Integer	Unique numeric code identifying the state
NO_UF	Geographical	String	Full name of the state
SG_UF	Geographical	String	Two-letter abbreviation of the state
CO_MUNICIPIO	Geographical	Integer	Unique numeric code identifying the municipality
NO_MUNICIPIO	Geographical	String	Full name of the municipality
TP_DEPENDENCIA	Administrative	Integer	School dependency type (1=Federal, 2=State, 3=Municipal, 4=Private)
TP_LOCALIZACAO	Administrative	Integer	School location type (1=Urban, 2=Rural)
IN_AGUA_POTAVEL	Infrastructure	Binary (0/1)	Indicates whether the school has access to potable water
IN_BANHEIRO	Infrastructure	Binary (0/1)	Indicates whether the school has bathroom

			facilities
IN_BIBLIOTECA	Infrastructure	Binary (0/1)	Indicates whether the school has a library
IN_COMPUTADOR	Infrastructure	Binary (0/1)	Indicates whether the school has computers
IN_ENERGIA_INEXISTENTE	Infrastructure	Binary (0/1)	Indicates whether the school has no electricity supply
IN_ESGOTO_INEXISTENTE	Infrastructure	Binary (0/1)	Indicates whether the school has no sewage system
IN_INTERNET	Infrastructure	Binary (0/1)	Indicates whether the school has internet access
IN_REFEITORIO	Infrastructure	Binary (0/1)	Indicates whether the school has a canteen or dining facility
NU_ANO_CENSO	Temporal	Integer	The census year in which the record was collected (2010–2020)
QT_MAT_BAS	Enrolment Metric	Integer	Total number of basic education enrolments per school per year

2.3 Data Quality and Issues

Upon initial exploration of the raw CSV files, several data quality issues were identified. The files use a semicolon (;) as the field delimiter rather than a comma, and are encoded in Latin-1 (ISO-8859-1) rather than UTF-8 — common characteristics of Brazilian government datasets requiring explicit handling during ingestion. Certain fields also contain missing or null values, particularly in infrastructure indicator columns where schools may not have reported specific attributes. Given the volume of over 2.7 million final fact records, efficient handling of these issues was essential to maintain pipeline performance.

2.4 Relational Data Model

The relational data model reflects the structure of the raw Censo Escolar data before transformation into the analytical Star Schema. At the centre is the school entity, identified by a unique school code, with relationships to its geographical location, administrative classification, and annual enrolment and infrastructure records. The location entity captures the hierarchical relationship between municipalities and states, with each school belonging to exactly one municipality and each municipality belonging to exactly one state.

3.0 Data Preparation

3.1 Data Extraction

The first step involved extracting raw CSV files from the Brazilian Open Data portal using a Python script (`download_censo.py`) that downloads twelve annual files covering 2010 to 2021. Each file was stored locally in a dedicated data directory. This approach ensures reproducibility, as the download script can be re-executed to refresh the dataset whenever new Censo Escolar data is published. The downloaded files are named consistently by year, allowing subsequent processing steps to iterate over them in a predictable sequence.

3.2 CSV to Parquet Conversion

Once the raw CSV files were downloaded, the next step was converting them from CSV to Apache Parquet format using PySpark (`convert_to_parquet.py`). This conversion is a critical performance optimization because Parquet is a columnar storage format that is significantly more compact and faster to query than CSV, particularly for analytical workloads accessing only a subset of columns. Each CSV file was read using Spark's DataFrameReader with:

- Header option set to `true`
- Delimiter set to semicolon (`;`)
- Encoding set to Latin-1
- Schema inference disabled for performance

The resulting DataFrames were written to disk in Parquet format using overwrite mode, producing files approximately five times smaller than the original CSVs.

3.3 Handling Missing Values and Inconsistencies

Missing values in infrastructure indicator columns were handled by treating null entries as 0, indicating the absence of a facility or resource — consistent with the binary semantics of these variables. Column data types were cast explicitly where necessary to ensure consistency across years. String columns containing state and municipality names were retained as-is, while numeric columns such as school codes and enrolment counts were cast to long integer types.

4.0 Modeling

4.1 Star Schema Design

The modeling phase centred on designing and implementing a multidimensional Star Schema — the industry-standard data architecture for analytical and business intelligence workloads. The Star Schema organises data into a central fact table surrounded by multiple dimension tables, each connected via foreign key references. This structure significantly accelerates aggregation queries by reducing complex joins during reporting, which is particularly beneficial when querying nearly three million records.

The Star Schema implemented in this project consists of:

- **1 Fact Table:** `fact_censo_escolar` — contains total enrolments per school per year with foreign keys to all dimension tables
- **10 Dimension Tables:**
    - `dim_local` — geographical information (municipality and state)
    - `dim_tp_dependencia` — administrative dependency type
    - `dim_tp_localizacao` — urban or rural classification
    - `dim_in_agua_potavel` — potable water availability
    - `dim_in_banheiro` — bathroom facilities
    - `dim_in_biblioteca` — library availability
    - `dim_in_computador` — computer availability
    - `dim_in_energia_inexistente` — electricity absence indicator
    - `dim_in_esgoto_inexistente` — sewage absence indicator
    - `dim_in_internet` — internet access
    - `dim_in_refeitorio` — canteen availability

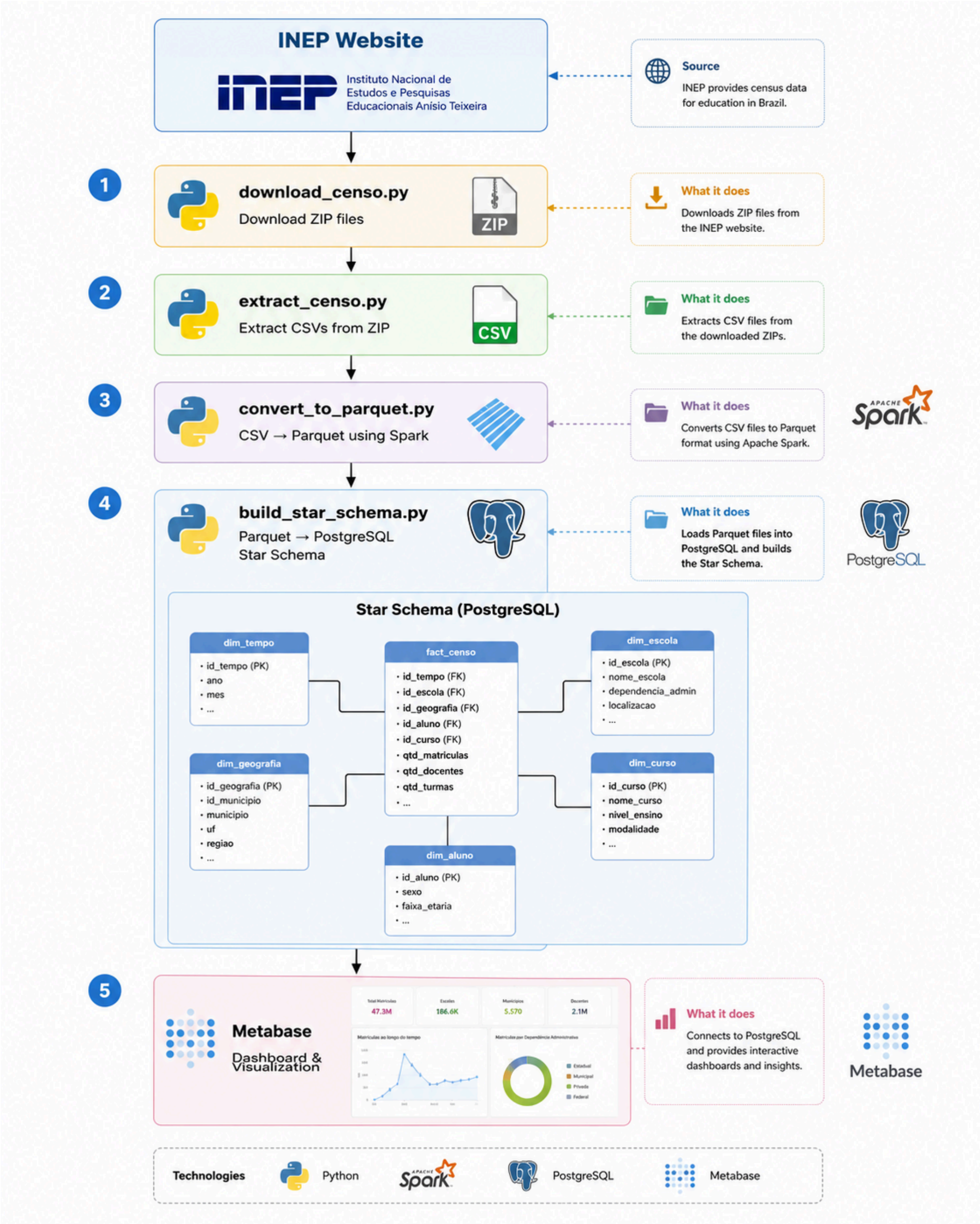
4.2 Implementation with Apache Spark and PostgreSQL

The Star Schema was implemented using PySpark for data transformation and JDBC for database interaction with PostgreSQL. For each dimension table:

1. Relevant columns were selected from the Parquet DataFrame
2. Duplicates were removed using Spark's `dropDuplicates()` transformation
3. A unique surrogate key was assigned using `monotonically_increasing_id()`
4. The resulting DataFrames were written to PostgreSQL using Spark's JDBC connector

The fact table was constructed through a series of left join operations between the main DataFrame and each dimension table, matching on natural key columns. The complete pipeline produced a fact table containing **2,792,984 records**, confirming successful end-to-end execution.

4.3 Pipeline Architecture



The pipeline was orchestrated using Apache Spark 3.5.1 in a local Windows environment with Docker managing three supporting services: PostgreSQL as the data warehouse, Metabase for dashboard visualisation, and Adminer for database administration.

5.0 Evaluation

5.1 Pipeline Performance

From a technical perspective, the pipeline successfully processed all 12 years of data with the following results:

- **2,792,984 records** loaded into the fact table
- **~80% file size reduction** achieved through CSV to Parquet conversion
- **11 dimension tables** created with correct surrogate keys
- All foreign key references resolved without null mismatches
- Pipeline completed without data loss or schema inconsistencies

5.2 Business Utility and Dashboard Results

The Star Schema enabled the construction of a Metabase dashboard answering key analytical questions about Brazilian basic education. Key findings include:

- Total enrolments peaked at approximately **52 million in 2010** and declined steadily to **46.7 million by 2021**, reflecting demographic and policy changes in Brazil's education system
- **São Paulo** leads all cities with approximately 2,674,792 students enrolled in 2021, followed by **Rio de Janeiro** (1,279,679) and **Brasília** (644,293)
- **67.5% of schools** had internet access in 2021, while **32.5% still lacked connectivity** — highlighting a significant digital divide in Brazil's education system
- São Paulo state dominates enrolment figures at the state level, with Minas Gerais and Bahia ranking second and third respectively

5.3 Limitations and Future Improvements

Despite successful execution, several limitations were identified:

- **Schema inference** during CSV loading requires a full file scan, adding processing time — a predefined schema would improve performance
- **Batch processing** requires reloading the entire dataset for any correction — incremental loading with year-based partitioning would significantly improve efficiency
- **Dashboard coverage** is currently limited to enrolment counts and geographical breakdowns — incorporating infrastructure quality trends over time would provide richer insights
- **No machine learning** component — future work could explore predicting future enrolment trends based on demographic and socioeconomic variables

6.0 Reflection

Throughout the execution of this project, I encountered several technical challenges that significantly contributed to my growth as a data engineer. The most demanding aspect was configuring the Spark environment on Windows, particularly resolving compatibility issues between Python 3.12 and PySpark 3.5.1, which required downgrading to Python 3.11 and creating a dedicated virtual environment. I also faced challenges with missing Hadoop native libraries on Windows, which was resolved by downloading the correct `winutils.exe` and `hadoop.dll` files for Hadoop 3.3.5.

In terms of skills gained, this project gave me hands-on experience with PySpark that I had never had before, especially in reading and writing Parquet files and performing complex DataFrame transformations such as `dropDuplicates`, `join`, and `monotonically_increasing_id` for surrogate key generation. I also developed a much deeper understanding of the Star Schema design, particularly why dimension tables are separated from the fact table and how this separation dramatically speeds up aggregation queries in business intelligence tools like Metabase. This project demonstrated that real-world data engineering requires not just programming skills, but also systematic debugging, environment management, and the ability to adapt when tools behave differently across operating systems.